

Design and Closed-loop Control of a Boost PFC Rectifier

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Specifications

- AC input voltage = 230 Vrms, 50 Hz, single-phase
- DC Output voltage = 400 V
- Rated output power = 1.2 kW
- Minimum output power = 0.24 kW
- Switching frequency = 500 kHz
- Input current THD < 5%



Assignment - A: DC-DC Boost converter analysis and design

1.a) The circuit diagram of DC- DC Boost Converter is given by :-

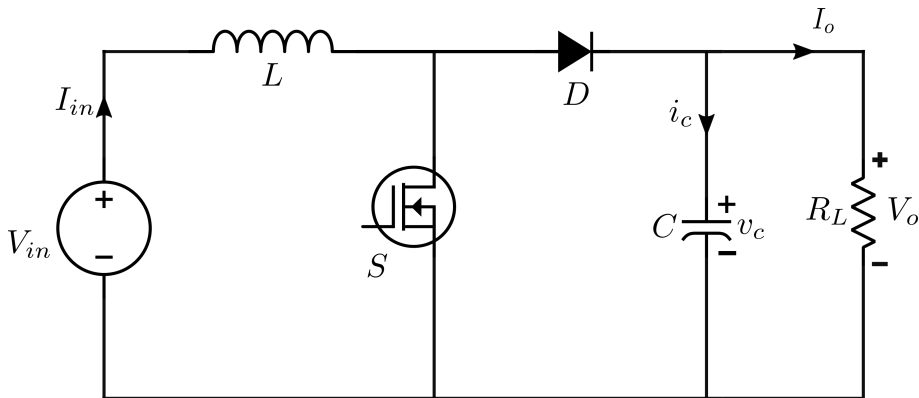


Figure 1: DC-DC Boost Converter



DC-DC Boost converter analysis and design

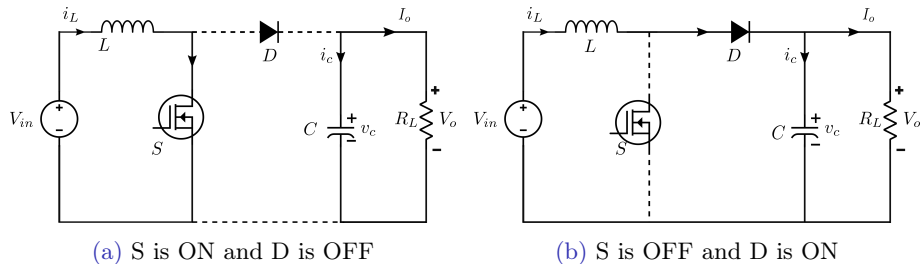


Figure 2: Circuit states depending on state of MOSFET



Derivations and Waveforms

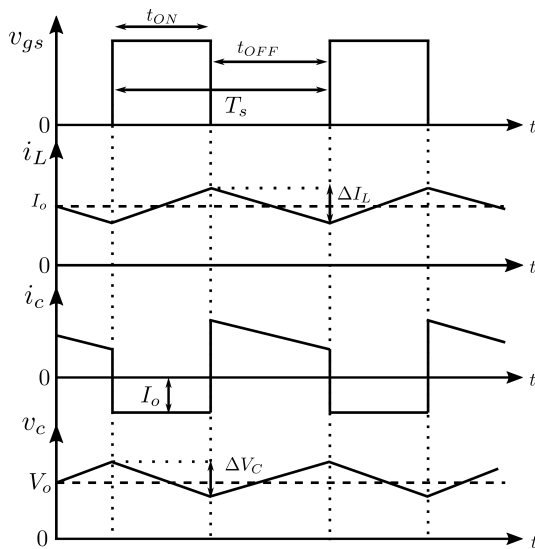


Figure 3: Waveforms of Inductor current, capacitor voltage and current



Derivations and Waveforms

The Inductor voltage expression for CCM is given by:-

$$\int_0^{T_s} v_L dt = 0 \Rightarrow \int_0^{t_{ON}} V_{in} dt + \int_{t_{ON}}^{T_s} (V_{in} - V_o) dt = 0$$

$$\Rightarrow V_{in} \cdot t_{ON} + (V_{in} - V_o) \cdot (T_s - t_{ON}) = 0$$

Defining Duty ratio,

$$D = \frac{t_{ON}}{T_s}$$

and

$$\text{Voltage Gain} = \frac{V_o}{V_{in}}$$

Dividing by T_s (Switching period) and rearranging we get,

$$\boxed{\frac{V_o}{V_{in}} = \frac{1}{1 - D}}$$



Derivations and Waveforms

As seen from the plot, Voltage gain (on p.u.scale) rises steeply with Duty ratio as it goes to 1

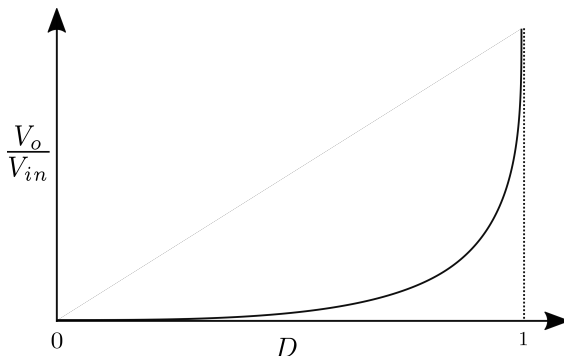


Figure 4: Voltage Gain vs Duty ratio



Derivations and Waveforms

1.b) The pk-pk inductor current ripple is given by

$$\Delta I_L = I_{\max} - I_{\min}$$

$$v_L(t) = L \frac{di(t)}{dt} \Rightarrow \int_{I_{\min}}^{I_{\max}} di = \int_0^{t_{\text{ON}}} \frac{v_L(t)}{L} dt = \frac{V_{in}}{L} \int_0^{t_{\text{ON}}} dt$$

$$\Rightarrow I_{\max} - I_{\min} = \frac{V_{in}}{L} t_{\text{ON}} \quad [\text{since } v_L \text{ during } t_{\text{ON}} \text{ is } V_{in} \text{ from } v_L \text{ waveform}]$$

$$\Rightarrow \Delta I_L = \frac{V_{in}}{L} (DT_s) \quad \left(\because \frac{t_{\text{ON}}}{T_s} = D \Rightarrow t_{\text{ON}} = DT_s \right)$$

$$\Rightarrow \Delta I_L = \frac{V_{in} D}{L f_s} \quad \left(\because T_s = \frac{1}{f_s} \right) \quad (2)$$

$$\Delta I_L = \frac{V_o D(1 - D)}{L f_s}$$



Derivations and Waveforms

Taking V_o , D and f_s to be constant, ΔI_L vs L can be plotted as,

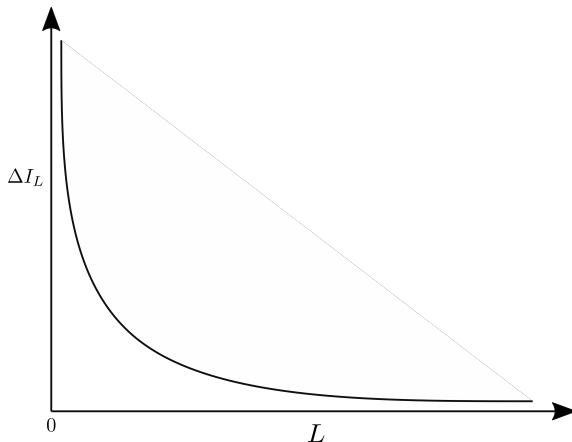


Figure 5: Pk-Pk Inductor Current ripple vs Inductance



Derivations and Waveforms

Using the same equation, taking L , f_s and V_o to be constant, ΔI_L vs D can be plotted as,

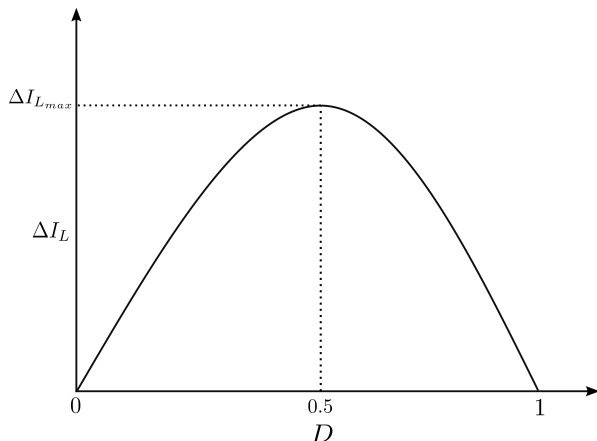


Figure 6: Pk-Pk Inductor Current ripple vs duty ratio



1.c) The capacitor current i_C can be given by,

$$i_C = C \frac{dv_C(t)}{dt} \Rightarrow \int_0^{\Delta V_C} dv_C(t) = \frac{1}{C} \int_0^{t_{\text{ON}}} i_C dt \quad (\because i_C(t) = I_o)$$

$$\Rightarrow \Delta V_C = \frac{1}{C} I_o t_{\text{ON}}$$

$$\Delta V_C = \frac{I_o D T_s}{C} = \frac{V_o D}{R C f_s} \quad \left(\because I_o = \frac{V_o}{R} \text{ \& } T_s = \frac{1}{f_s} \right) \quad (3)$$

$$\Delta V_C = \frac{V_{in} D T_s}{R C (1 - D)} \quad \left(\because \frac{V_o}{V_{in}} = \frac{1}{1 - D} \Rightarrow V_o = \frac{V_{in}}{1 - D} \right)$$

$$\Delta V_C = \frac{V_{in}}{R f_s} \cdot \frac{1}{C} \cdot \frac{D}{1 - D}$$



Derivations and Waveforms

For a constant V_{in} , R , f_s and D , capacitor voltage ripple, ΔV_C vs capacitance C can be given by:

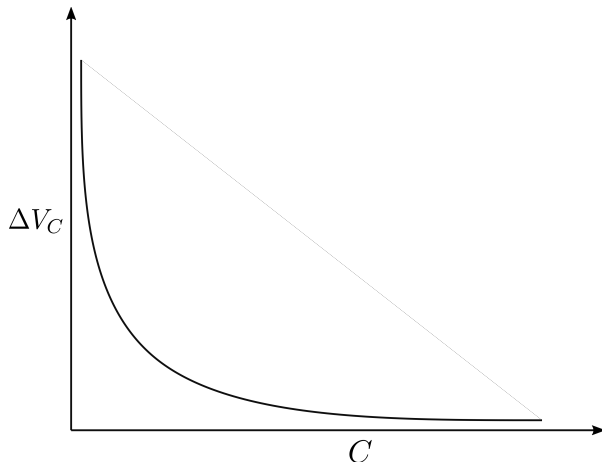


Figure 7: Pk-Pk Capacitor Voltage ripple vs Capacitance



Derivations and Waveforms

Keeping other parameters constant, ΔV_C vs D can be given as,

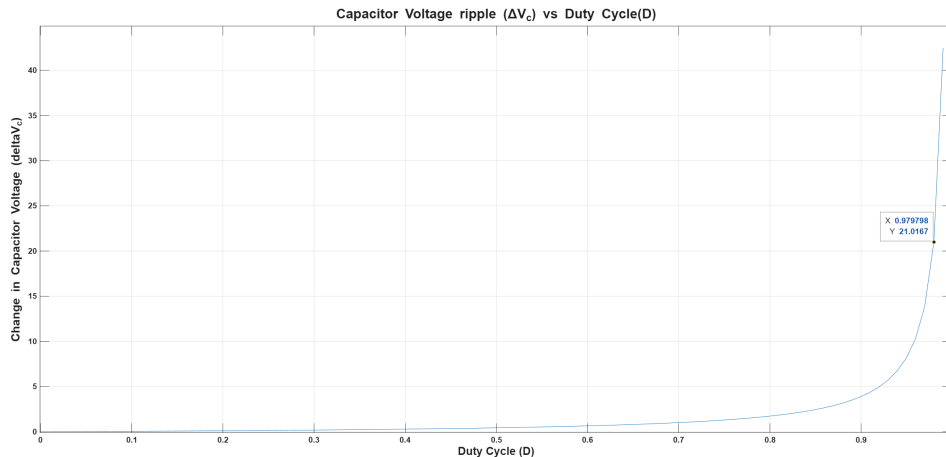


Figure 8: Pk-Pk Capacitor Voltage ripple vs Duty ratio



Design of DC-DC boost converter

Design Specifications

- Input voltage range = 10 V – 325 V
- Output voltage = 400 V
- Output power range = 1.2 kW – 0.24 kW

Constraints

- Worst-case pk-pk inductor current ripple = 60 % of the rated average current
- Worst-case pk-pk capacitor voltage ripple = 0.5% of the DC output voltage



Design of DC-DC Boost Converter

To design the Boost Converter, we need to select L and C value for a given operating point, and to do that we implement the formulae we developed for pk-pk inductor current ripple and pk-pk capacitor voltage ripple. From (2),

$$\Delta I_L = \frac{V_{in} D}{L f_s} = \frac{V_{in} \left(1 - \frac{V_{in}}{V_o}\right)}{L f_s} \left(\because \frac{V_o}{V_{in}} = \frac{1}{1-D} \Rightarrow D = 1 - \frac{V_{in}}{V_o} \right)$$

For average output power (P_o) 1.2 kW, the average output current is:

$$I_o = \frac{P_o}{V_o} = \frac{1200}{400} = 3A$$

For worst case pk-pk inductor current ripple to be 60% of rated average current

$$\Rightarrow \Delta I_L = 0.6 \times 3 = 1.8A.$$



Calculation of Inductance (L)

For $V_{in}=10$ V, putting into the above formula to find Inductance(L),

$$L = \frac{V_{in}(1 - \frac{V_{in}}{V_o})}{\Delta I_L f_s} = \frac{10(1 - \frac{10}{400})}{1.8 \times 500 \times 10^3} = 10.833 \mu\text{H}$$

Output Resistance(R_o),

$$R_o = \frac{V_o^2}{P_o} = \frac{400^2}{1200} = 133.3 \Omega$$

To find Capacitance(C), pk-pk capacitor voltage ΔV_c is taken to be 0.5% of output voltage ie.

$$\Delta V_c = 0.005 \times 400 = 2V$$

From the expression of capacitor voltage, from (3),

$$\Delta V_C = \frac{V_o D}{RC f_s} = \frac{V_o}{RC f_s} \left(1 - \frac{V_{in}}{V_o}\right)$$



Calculation of Capacitance(C)

Replacing output voltage V_o and R by rated output current I_o , we get

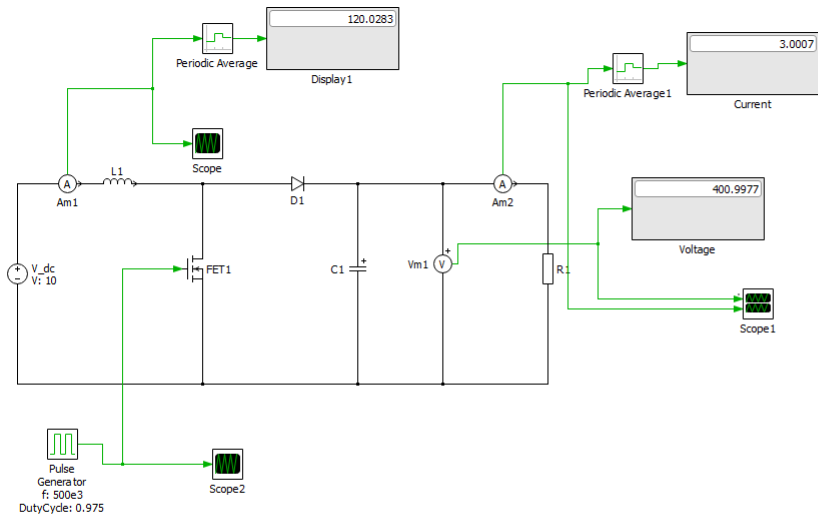
$$\Rightarrow \Delta V_C = \frac{I_o}{C f_s} \left(1 - \frac{V_{in}}{V_o} \right)$$

$$\Rightarrow C = \frac{I_o}{\Delta V_c f_s} \left(1 - \frac{V_{in}}{V_o} \right)$$

$$C = \frac{I_o D}{\Delta V_c f_s} = \frac{3 \times 0.975}{2 \times 500 \times 10^3} = 2.925 \mu\text{F} \left(\because D = 1 - \frac{10}{400} = 0.975 \right)$$



PLECS Simulation(IDEAL)



MATLAB Script for calculation across various operating points

```
BoostCalculation.m x +
C:\Users\PC\Desktop\Matlab scripts\BoostCalculation.m
1 % Parameters
2 Vin_vals = [10, 325];
3 Po_vals = [240, 1200];
4 Vo = 400; % Output voltage (V)
5 fs = 500e3; % Switching frequency (Hz)
6 delta_IL_percent = 0.6;
7 delta_Vc_percent = 0.005;
8 % Initialize result table
9 results = [];
10 % Loop over base conditions
11 for Vin_base = Vin_vals
12     for Po_base = Po_vals
13         Iout_base = Po_base / Vo;
14         D_base = 1 - Vin_base / Vo;
15         delta_IL_base = delta_IL_percent * Iout_base;
16         delta_Vc_spec = delta_Vc_percent * Vo;
17
18         % Calculate L and C for base point
19         L_base = (Vin_base * D_base) / (fs * delta_IL_base);
20         C_base = (Iout_base * D_base) / (fs * delta_Vc_spec);
21         % Loop over all eval conditions
22         for Vin_eval = Vin_vals
23             for Po_eval = Po_vals
24                 Iout_eval = Po_eval / Vo;
25                 D_eval = 1 - Vin_eval / Vo;
26
27                 % Recalculate ripple values using base L and C at eval point
28                 delta_IL_eval = (Vin_eval * D_eval) / (fs * L_base);
29                 delta_Vc_eval = (Iout_eval * D_eval) / (fs * C_base);
30
31                 % Store result row
32                 results = [results;
33                     Vin_base, Po_base, D_base, L_base*1e6, C_base*1e6, ...
34                     Vin_eval, Po_eval, D_eval, delta_IL_eval, delta_Vc_eval];
35             end
36         end
37     end
38 end
39
40 % Create table with headers
41 result_table = array2table(results, ...
42     'VariableNames', {'Base_Vin', 'Base_Po', 'Base_D', 'L_uH', 'C_uF', ...
43     'Eval_Vin', 'Eval_Po', 'Eval_D', 'Delta_IL_A', 'Delta_Vc_V'});
44 % Display the table
45 disp(result_table);
46
```

Figure 10: MATLAB Code



MATLAB Results

BoostCalculation.m									
Command Window									
>> BoostCalculation									
Base_Vin	Base_Po	Base_D	L_uH	C_uF	Eval_Vin	Eval_Po	Eval_D	Delta_IL_A	Delta_Vc_V
10	240	0.975	54.167	0.585	10	240	0.975	0.36	2
10	240	0.975	54.167	0.585	10	1200	0.975	0.36	10
10	240	0.975	54.167	0.585	325	240	0.1875	2.25	0.38462
10	240	0.975	54.167	0.585	325	1200	0.1875	2.25	1.9231
10	1200	0.975	10.833	2.925	10	240	0.975	1.8	0.4
10	1200	0.975	10.833	2.925	10	1200	0.975	1.8	2
10	1200	0.975	10.833	2.925	325	240	0.1875	11.25	0.076923
10	1200	0.975	10.833	2.925	325	1200	0.1875	11.25	0.38462
325	240	0.1875	338.54	0.1125	10	240	0.975	0.0576	10.4
325	240	0.1875	338.54	0.1125	10	1200	0.975	0.0576	52
325	240	0.1875	338.54	0.1125	325	240	0.1875	0.36	2
325	240	0.1875	338.54	0.1125	325	1200	0.1875	0.36	10
325	1200	0.1875	67.708	0.5625	10	240	0.975	0.288	2.08
325	1200	0.1875	67.708	0.5625	10	1200	0.975	0.288	10.4
325	1200	0.1875	67.708	0.5625	325	240	0.1875	1.8	0.4
325	1200	0.1875	67.708	0.5625	325	1200	0.1875	1.8	2

Figure 11: MATLAB Results



Design 1 : $L = 54.17 \mu\text{H}$, $C = 0.58 \mu\text{F}$

Table 1: Base operating point: $V_{in}=10\text{V}$, $P_o=240\text{W}$

Operating Point	Ripple	Analysis	Simulation	Error(%)
$V_{in}=10\text{V}$, $P_o=240\text{W}$, $D=0.975$	$\Delta I_L(A)$	0.360	0.36	0
	$\Delta V_C(V)$	2.000	2.00	0
$V_{in}=10\text{V}$, $P_o=1200\text{W}$, $D=0.975$	$\Delta I_L(A)$	0.360	0.36	0
	$\Delta V_C(V)$	10.000	10	0
$V_{in}=325\text{V}$, $P_o=240\text{W}$, $D=0.188$	$\Delta I_L(A)$	2.250	2.25	0
	$\Delta V_C(V)$	0.385	1(DCM)	61.5
$V_{in}=325\text{V}$, $P_o=1200\text{W}$, $D=0.188$	$\Delta I_L(A)$	2.250	2.24	0.44
	$\Delta V_C(V)$	1.923	1.95	1.4

Error is calculated as:

$$\frac{\text{Simulation} - \text{Analysis}}{\text{Simulation}} \times 100\%$$



Design 2 : $L = 10.83 \mu\text{H}$, $C = 2.92 \mu\text{F}$

Table 2: Base operating point: $V_{in}=10\text{V}$, $P_o=1200\text{W}$

Operating Point	Ripple	Analysis	Simulation	Error (%)
$V_{in}=10\text{V}$, $P_o=240\text{W}$, $D=0.975$	$\Delta I_L(A)$	1.800	1.8	0
	$\Delta V_C(V)$	0.400	0.4	0
$V_{in}=10\text{V}$, $P_o=1200\text{W}$, $D=0.975$	$\Delta I_L(A)$	1.800	1.8	0
	$\Delta V_C(V)$	2.000	2.0	0
$V_{in}=325\text{V}$, $P_o=240\text{W}$, $D=0.188$	$\Delta I_L(A)$	11.250	11.28(DCM)	0.266
	$\Delta V_C(V)$	0.077	0.824	90.65
$V_{in}=325\text{V}$, $P_o=1200\text{W}$, $D=0.188$	$\Delta I_L(A)$	11.250	11.27(DCM)	0.18
	$\Delta V_C(V)$	0.385	1	61.5



Design 3: $L = 338.54 \mu\text{H}$, $C = 0.11 \mu\text{F}$

Table 3: Base operating point: $V_{\text{in}}=325\text{V}$, $P_o=240\text{W}$

Operating Point	Ripple	Analysis	Simulation	Error(%)
$V_{\text{in}}=10\text{V}$, $P_o=240\text{W}$, $D=0.975$	$\Delta I_L(A)$	0.058	0.06	3.33
	$\Delta V_C(V)$	10.400	10.63	2.16
$V_{\text{in}}=10\text{V}$, $P_o=1200\text{W}$, $D=0.975$	$\Delta I_L(A)$	0.058	0.06	0.34
	$\Delta V_C(V)$	52.000	53.12	2.10
$V_{\text{in}}=325\text{V}$, $P_o=240\text{W}$, $D=0.188$	$\Delta I_L(A)$	0.360	0.36	0
	$\Delta V_C(V)$	2.000	2	0
$V_{\text{in}}=325\text{V}$, $P_o=1200\text{W}$, $D=0.188$	$\Delta I_L(A)$	0.360	0.36	0
	$\Delta V_C(V)$	10.000	10.22	2.15



Design 4: $L = 67.71 \mu\text{H}$, $C = 0.56 \mu\text{F}$

Table 4: Base operating point: $V_{\text{in}}=325\text{V}$, $P_o=1200\text{W}$

Operating Point	Ripple	Analysis	Simulation	Error(%)
$V_{\text{in}}=10\text{V}$, $P_o=240\text{W}$, $D=0.975$	$\Delta I_L(A)$	0.288	0.29	0.7
	$\Delta V_C(V)$	2.080	2.1	0.96
$V_{\text{in}}=10\text{V}$, $P_o=1200\text{W}$, $D=0.975$	$\Delta I_L(A)$	0.288	0.3	0.7
	$\Delta V_C(V)$	10.400	10.4	0
$V_{\text{in}}=325\text{V}$, $P_o=240\text{W}$, $D=0.188$	$\Delta I_L(A)$	1.800	1.8	0.0
	$\Delta V_C(V)$	0.400	0.87	54
$V_{\text{in}}=325\text{V}$, $P_o=1200\text{W}$, $D=0.188$	$\Delta I_L(A)$	1.800	1.8	0
	$\Delta V_C(V)$	2.000	2	0



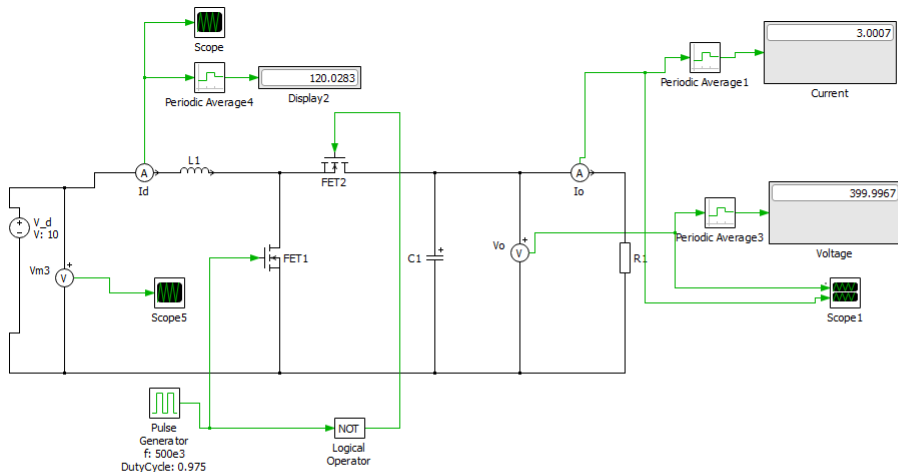
Design 5: $L = 111.1 \mu\text{H}$, $C = 1.5 \mu\text{F}$

Table 5: Base: $V_{in}=325\text{V}$, $P_o=1200\text{W}$

Operating Point	Ripple	Analysis	Simulation	Error(%)
$V_{in}=10\text{V}$, $P_o=240\text{W}$, $D=0.975$	$\Delta I_L(A)$	0.176	0.17	3.4
	$\Delta V_C(V)$	0.78	0.75	3.85
$V_{in}=10\text{V}$, $P_o=1200\text{W}$, $D=0.975$	$\Delta I_L(A)$	0.176	0.17	3.4
	$\Delta V_C(V)$	3.9	3.9	0
$V_{in}=325\text{V}$, $P_o=240\text{W}$, $D=0.188$	$\Delta I_L(A)$	1.1	1.1	0
	$\Delta V_C(V)$	0.15	0.3	50
$V_{in}=325\text{V}$, $P_o=1200\text{W}$, $D=0.188$	$\Delta I_L(A)$	1.1	1.1	0
	$\Delta V_C(V)$	0.75	0.7	6.66



Boost Converter with Synchronous MOSFET



Voltage Ripple ΔV_c : CCM vs DCM

Assumptions:

- Output voltage V_o is regulated
- Output current $I_{out} = \frac{P_o}{V_o}$
- Switching frequency $f_s = \frac{1}{T_s}$, Capacitance C

CCM (Continuous Conduction Mode):

- Capacitor supplies load only for duration $D \cdot T_s$
- Ripple derived from $\Delta Q = I_{out} \cdot D \cdot T_s$

$$\Delta V_c^{\text{CCM}} = \frac{\Delta Q}{C} = \frac{I_{out} \cdot D}{f_s \cdot C}$$

DCM (Discontinuous Conduction Mode):

- Inductor stops conducting; capacitor supplies load for nearly full T_s
- Ripple derived from $\Delta Q = I_{out} \cdot T_s$

$$\Delta V_c^{\text{DCM}} = \frac{I_{out} \cdot T_s}{C} = \frac{I_{out}}{f_s \cdot C}$$



Capacitance Calculation

Table 6: Calculated Capacitance for CCM (with Diode) and DCM (Synchronous FET)

V_{in} (V)	P_o (W)	I_o (A)	D	C_{CCM} (μF)	C_{DCM} (μF)
10	240	0.6	0.975	0.585	0.300
10	1200	3.0	0.975	2.925	1.500
325	240	0.6	0.188	0.113	0.300
325	1200	3.0	0.188	0.564	1.500



Table 7: Calculation for Choosing Inductance Value

V _{in} (V)	P _o (W)	D	L (μH)	I _{peak} (A)	I _{rms} (A)	L · I _{peak} · I _{rms}
10	240	0.975	54.17	24.18	24.00	0.03143
10	1200	0.975	10.83	120.90	120.00	0.15717
325	240	0.188	338.54	0.918	0.746	0.00023
325	1200	0.188	67.71	4.592	3.729	0.00116
200	1200	0.500	111.10	6.900	6.022	0.00462

Metric of Volume = $L \cdot I_{\text{peak}} \cdot I_{\text{rms}}$

Current Formulas:

$$I_{\text{peak}} = I_{\text{avg}} + \frac{\Delta I_L}{2} \quad I_{\text{rms}} = \sqrt{I_{\text{avg}}^2 + \frac{(\Delta I_L)^2}{12}}$$



Current Analysis for Boost Converter Designs

Table 8: Current Analysis for Boost Converter Designs

V_{in} (V)	P_o (W)	D	I_{out} (A)	$I_{L,rms}$ (A)	$I_{sw1,rms}$ (A)	$I_{sw2,rms}$ (A)
10	240	0.975	0.6	24.00	23.72	0.135
10	1200	0.975	3.0	120.00	118.60	0.673
325	240	0.188	0.6	0.746	0.513	0.545
325	1200	0.188	3.0	3.729	2.564	2.726
200	1200	0.5	3.0	6.022	4.257	2.121

Key Equations:

$$I_{L,avg} = \frac{I_{out}}{1-D}$$

$$\Delta I_L = \frac{V_{in} \cdot D}{L \cdot f_s}$$

$$I_{L,pk} = I_{L,avg} + \frac{\Delta I_L}{2}$$

$$I_{L,RMS} = \sqrt{I_{L,avg}^2 + \frac{1}{12} (\Delta I_L)^2}$$

$$I_{S1,RMS} = I_{L,RMS} \cdot \sqrt{D}$$

$$I_{S2,RMS} = I_{L,RMS} \cdot \sqrt{1-D}$$



Conduction Losses in Synchronous Boost Converter

Table 9: Conduction Losses at Various Operating Points

Operating Point	$I_{L,\text{RMS}}$ (A)	P_{cond} (W)
$V_{\text{in}} = 10 \text{ V}, P_o = 240 \text{ W}$	24.00	2.88
$V_{\text{in}} = 10 \text{ V}, P_o = 1200 \text{ W}$	120.00	72.00
$V_{\text{in}} = 325 \text{ V}, P_o = 240 \text{ W}$	0.746	0.00278
$V_{\text{in}} = 325 \text{ V}, P_o = 1200 \text{ W}$	3.729	0.0695
$V_{\text{in}} = 200 \text{ V}, P_o = 1200 \text{ W}$	6.022	0.181

Formula used:

$$P_{\text{cond}} = I_{\text{RMS}}^2 \times R_{\text{DS(on)}}$$

Assumed: $R_{\text{DS(on)}} = 5 \text{ m}\Omega$



Boost Rectifier with AC Input and Synchronous MOSFET

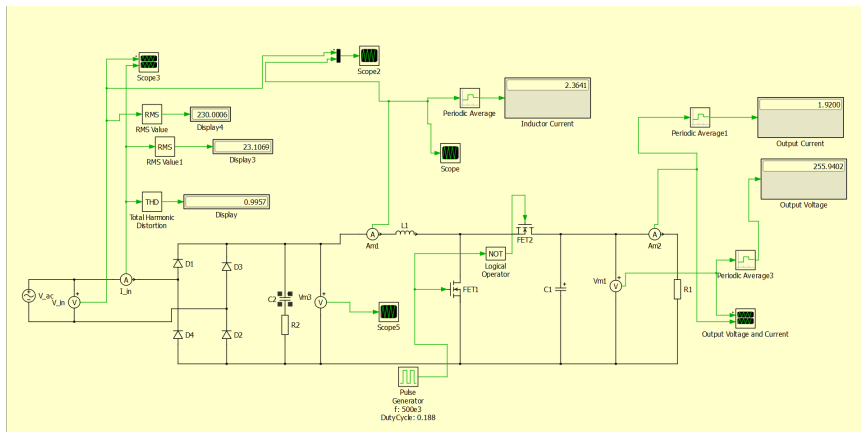


Figure 13: Boost Rectifier with AC input



Observations

- Inductor current ripple at zero crossing of AC Input = 0 A
- Inductor current ripple at peak of AC input = 1.1 A

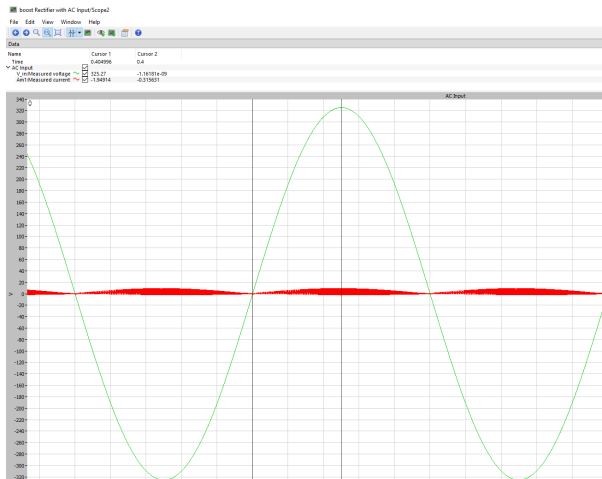


Figure 14: Inductor current ripple at zero and pk of AC input



- Input current THD : 0.0952

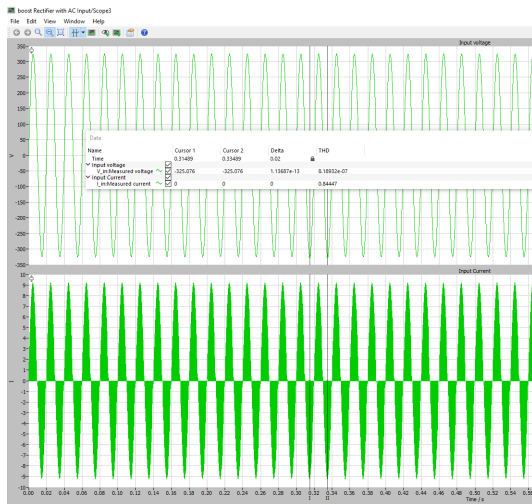


Figure 15: Input Current THD



Derivation of Large-Signal State-Space Model

State variables:

$$x_1 = i_L(t), \quad x_2 = v_o(t)$$

Mode 1: Switch ON (Diode OFF)

$$\dot{i}_L = \frac{V_{in}}{L}, \quad \dot{v}_o = -\frac{v_o}{RC}$$

$$\begin{bmatrix} \dot{i}_L \\ \dot{v}_o \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ v_o \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} V_{in}$$

Mode 2: Switch OFF (Diode ON)

$$\dot{i}_L = \frac{V_{in} - v_o}{L}, \quad \dot{v}_o = \frac{i_L - \frac{v_o}{R}}{C}$$

$$\begin{bmatrix} \dot{i}_L \\ \dot{v}_o \end{bmatrix} = \begin{bmatrix} 0 & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ v_o \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} V_{in}$$



Averaging over duty cycle $d(t)$:

$$A(d) = dA_1 + (1 - d)A_2, \quad B(d) = dB_1 + (1 - d)B_2$$

$$\begin{bmatrix} \dot{i}_L \\ \dot{v}_o \end{bmatrix} = \begin{bmatrix} 0 & -\frac{(1-d)}{L} \\ \frac{(1-d)}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ v_o \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} V_{in}$$



Perturb around steady-state:

$$i_L(t) = I_L + \hat{i}_L(t), \quad v_o(t) = V_o + \hat{v}_o(t), \quad d(t) = D + \hat{d}(t)$$

Linearized system:

$$\begin{bmatrix} \dot{\hat{i}}_L \\ \dot{\hat{v}}_o \end{bmatrix} = A \begin{bmatrix} \hat{i}_L \\ \hat{v}_o \end{bmatrix} + B \hat{d}(t)$$

Where:

$$A = \begin{bmatrix} 0 & -\frac{(1-D)}{L} \\ \frac{(1-D)}{C} & -\frac{1}{RC} \end{bmatrix}, \quad B = \begin{bmatrix} \frac{V_o}{L} \\ -\frac{I_L}{C} \end{bmatrix}$$



Control-to-Output Small Signal Transfer Function

Transfer function from $\hat{d}(s) \rightarrow \hat{i}_L(s)$:

$$G_{id}(s) = \frac{\hat{i}_L(s)}{\hat{d}(s)} = C(sI - A)^{-1}B \quad \text{where } C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

Final expression:

$$G_{id}(s) = \frac{V_o s + \frac{I_L}{RC}}{LCs^2 + \frac{L}{R}s + (1 - D)^2}$$

This transfer function is used in average current-mode control design.



MATLAB Implementation of Control to Output Transfer function

Considering corner cases

- Input voltage range = 10 V – 325 V
- Output power range = 1.2 kW – 0.24 kW

We plot Bode plot in MATLAB, code for which is given below:

```
1 % Parameters
2 Vo = 480; % Output voltage (V)
3 L = 10e-6; % Inductance (H)
4 C = 34e-6; % Capacitance (F)
5
6 % Corner cases
7 Vin_vals = [10, 325]; % Input voltage range (V)
8 Po_vals = [1200, 240]; % Output power range (W)
9
10 figure;
11 hold on;
12 legendStrs = {};
13
14 for Vin = Vin_vals
15     for Po = Po_vals
16         % Calculate dependent parameters
17         D = 1 - Vin / Vo; % Duty cycle
18         R = Vo^2 / Po; % Load resistance
19         IL = Po / Vin; % Inductor current
20
21         % Transfer function numerator and denominator
22         num = [Vo, IL / (R * C)];
23         den = [L * C, L / R, (1 - D)^2];
24
25         % Transfer function H(s) = (Vo*s + 1/(RC)) / (LC*s^2 + L/R*s + (1-D)^2)
26         sys = tf(num, den);
27
28         % Plot Bode
29         bode(sys, [1e2, 1e4]); grid on;
30         legendStrs(end+1) = sprintf('Vin=%dV, Po=%dW', Vin, Po);
31     end
32 end
33
34 title('Control-to-Inductor Current Transfer Function H_id(s)');
35 legend(legendStrs);
36
```

Figure 16: MATLAB Code for Transfer Function plot



Bode plot

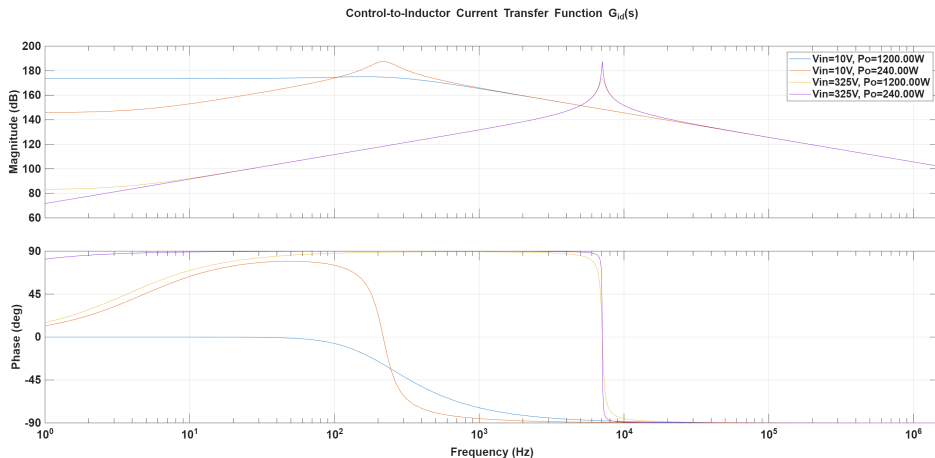


Figure 17: Bode plot for different corner cases

